Principles of Digital Communications
Project Theory

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Problem 1

Suppose

$$U = [U_1, \dots, U_n] \in \mathbb{R}^n$$

is uniformly distributed on the unit sphere

$$\{x \in \mathbb{R}^n : \|x\|^2 = 1\}.$$

A convenient construction is obtained by letting

$$X = [X_1, \dots, X_n] \sim \mathcal{N}(0, I_n)$$

and defining

$$U = \frac{X}{\|X\|}.$$

We now solve each part.

(a)

We want to show that for $-1 < t < 1$,

$$\Pr(U_1 \geq t) = \mathbb{E}\left[Q\left(\frac{t}{\sqrt{1-t^2}}\sqrt{X_2^2 + \cdots + X_n^2}\right)\right].$$

Since

$$U_1 = \frac{X_1}{\sqrt{X_1^2 + \cdots + X_n^2}},$$

the event $U_1 \geq t$ becomes

$$\frac{X_1}{\sqrt{X_1^2 + \cdots + X_n^2}} \geq t.$$

Let's define

$$S = X_2^2 + \cdots + X_n^2.$$

We distinguish two cases.

Case 1: $0 \leq t < 1$. Since the denominator is always positive, the inequality implies $X_1 \geq 0$, so squaring both sides is valid:

$$X_1^2 \geq t^2(X_1^2 + S).$$

Rearranging,

$$(1 - t^2)X_1^2 \geq t^2 S,$$

hence

$$X_1^2 \geq \frac{t^2}{1-t^2} S.$$

Since $X_1 \geq 0$,

$$X_1 \geq \frac{t}{\sqrt{1-t^2}}\sqrt{S}.$$

Conditioning on S,

$$\Pr(U_1 \geq t \mid S) = \Pr\left(X_1 \geq \frac{t}{\sqrt{1-t^2}}\sqrt{S}\right).$$

Since $X_1 \sim \mathcal{N}(0,1)$,

$$= Q\left(\frac{t}{\sqrt{1-t^2}}\sqrt{S}\right).$$

Case 2: $-1 < t < 0$**.** The inequality becomes

$$X_1 \geq t\sqrt{X_1^2 + S}.$$

If $X_1 \geq 0$, the inequality is automatically satisfied because the right-hand side is negative. If $X_1 < 0$, then both sides are negative and squaring is valid:

$$X_1^2 \leq t^2(X_1^2 + S).$$

Thus,

$$(1-t^2)X_1^2 \leq t^2 S,$$

which gives

$$|X_1| \leq \sqrt{\frac{t^2}{1-t^2}}\sqrt{S}.$$

Since $X_1 < 0$,

$$-\sqrt{\frac{t^2}{1-t^2}}\sqrt{S} \leq X_1 < 0.$$

Combining with the case $X_1 \geq 0$, we conclude that

$$U_1 \geq t \iff X_1 \geq -\sqrt{\frac{t^2}{1-t^2}}\sqrt{S}.$$

Since $t < 0$,

$$-\sqrt{\frac{t^2}{1-t^2}} = \frac{t}{\sqrt{1-t^2}}.$$

Therefore,

$$U_1 \geq t \iff X_1 \geq \frac{t}{\sqrt{1-t^2}}\sqrt{S}.$$

Conditioning on S,

$$\Pr(U_1 \geq t \mid S) = \Pr\left(X_1 \geq \frac{t}{\sqrt{1-t^2}}\sqrt{S}\right) = Q\left(\frac{t}{\sqrt{1-t^2}}\sqrt{S}\right).$$

Finally, taking expectation over S yields

$$\Pr(U_1 \geq t) = \mathbb{E}\left[Q\left(\frac{t}{\sqrt{1-t^2}}\sqrt{X_2^2 + \cdots + X_n^2}\right)\right].$$

(b)

We now show that for $0 \le t < 1$,

$$\Pr(U_1 \ge t) \le \frac{1}{2}(1-t^2)^{(n-1)/2}.$$

From part (a),

$$\Pr(U_1 \ge t) = \mathbb{E}\left[Q\left(\sqrt{\frac{t^2}{1-t^2}}\sqrt{X_2^2 + \cdots + X_n^2}\right)\right].$$

Using the bound

$$Q(z) \le \frac{1}{2}e^{-z^2/2},$$

we get

$$\Pr(U_1 \ge t) \le \frac{1}{2}\mathbb{E}\left[\exp\left(-\frac{1}{2}\frac{t^2}{1-t^2}(X_2^2 + \cdots + X_n^2)\right)\right].$$

Define

$$\lambda = \frac{t^2}{1-t^2}.$$

Then

$$\Pr(U_1 \ge t) \le \frac{1}{2}\mathbb{E}\left[e^{-\lambda(X_2^2+\cdots+X_n^2)/2}\right].$$

Since the random variables are independent,

$$= \frac{1}{2}\prod_{i=2}^{n}\mathbb{E}\left[e^{-\lambda X_i^2/2}\right].$$

Using the identity

$$\mathbb{E}[e^{-\lambda Z^2/2}] = \frac{1}{\sqrt{1+\lambda}},$$

for $Z \sim \mathcal{N}(0,1)$, we obtain

$$= \frac{1}{2}\left(\frac{1}{\sqrt{1+\lambda}}\right)^{n-1}.$$

Now,

$$1+\lambda = 1 + \frac{t^2}{1-t^2} = \frac{1}{1-t^2},$$

hence

$$\frac{1}{\sqrt{1+\lambda}} = \sqrt{1-t^2}.$$

Therefore,

$$\Pr(U_1 \ge t) \le \frac{1}{2}(1-t^2)^{(n-1)/2}.$$

(c)

Let $v \in \mathbb{R}^n$ be any unit vector. We show that

$$\langle U, v\rangle \stackrel{d}{=} U_1.$$

Since v is a unit vector, there exists an orthogonal matrix R such that

$$Rv = e_1,$$

where

$$e_1 = (1, 0, \dots, 0).$$

Because U is uniformly distributed on the sphere, its distribution is rotationally invariant. Therefore,

$$RU \stackrel{d}{=} U.$$

Now,

$$\langle U, v \rangle = \langle RU, Rv \rangle = \langle RU, e_1 \rangle.$$

But

$$\langle x, e_1 \rangle = x_1,$$

so

$$\langle RU, e_1 \rangle = (RU)_1.$$

Since $RU \stackrel{d}{=} U$,

$$(RU)_1 \stackrel{d}{=} U_1.$$

Hence,

$$\langle U, v \rangle \stackrel{d}{=} U_1.$$

Problem 2

Suppose the codewords $c_1, ..., c_m$ being vectors in $\mathbb{R}^n$ with $\|c_1\|^2 = \cdots = \|c_m\|^2 = E$, and the m messages being equally likely. For a received vector $y \in \mathbb{R}^n$

$$\text{score}(i, y) = \frac{\langle y, c_i \rangle}{\|c_i\| \|y\|} \in [-1, +1].$$

We now solve each part.

(a)

We want to show that the receiver deciding $\hat{\imath} = \arg\max_i \text{score}(i, y)$ minimizes the probability of error.

Over an AWGN channel, this correspond to the Maximum Likelihood (ML) receiver, which maximizes the conditional probability density function

$$f_{Y|X}(y \mid c_i) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{\|y - c_i\|^2}{2\sigma^2}\right).$$

Maximizing this density is equivalent to minimizing the squared Euclidean distance

$$\|y - c_i\|^2 = \|y\|^2 - 2\langle y, c_i \rangle + \|c_i\|^2.$$

Since $\forall i$, $\|c_i\|^2 = E$, and $\|y\|^2$ is independent of i, minimizing $\|y - c_i\|^2$ is strictly equivalent to maximizing the inner product $\langle y, c_i \rangle$,

$$\hat{\imath}_{\text{ML}} = \arg\max_i \langle y, c_i \rangle.$$

Since $\forall i$, $\|c_i\| = \sqrt{E}$, and $\|y\|$ ¿ 0,

$$\arg\max_i \text{score}(i, y) = \arg\max_i \frac{\langle y, c_i \rangle}{\sqrt{E}\|y\|} = \arg\max_i \langle y, c_i \rangle.$$

(b)

Suppose message 1 is sent so that $Y = c_1 + Z$, where $Z \sim \mathcal{N}(0, \sigma^2 I_n)$. Let $-1 < t < 1$. We want to show that

$$\Pr(\text{score}(1, Y) \leq t) = \Pr\left(\sqrt{E} + Z_1 \leq \frac{t}{\sqrt{1 - t^2}} \sqrt{Z_2^2 + \cdots + Z_n^2}\right).$$

By the rotational invariance of Z, we can a rotation to our coordinate system such that the first basis vector aligns with c_1. In this new coordinate system:

$$c_1 = (\sqrt{E}, 0, \ldots, 0)^T.$$

We still have $Z_i \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$. And

$$Y = (\sqrt{E} + Z_1, Z_2, \ldots, Z_n)^T.$$

Then

$$\langle Y, c_1 \rangle = \sqrt{E}(\sqrt{E} + Z_1)$$

And,

$$\|c_1\| = \sqrt{E}$$

Substituting,

$$\text{score}(1, Y) = \frac{\sqrt{E}(\sqrt{E} + Z_1)}{\sqrt{E}\sqrt{(\sqrt{E} + Z_1)^2 + Z_2^2 + \cdots + Z_n^2}} = \frac{\sqrt{E} + Z_1}{\sqrt{(\sqrt{E} + Z_1)^2 + Z_2^2 + \cdots + Z_n^2}}.$$

Then,

$$\begin{aligned} &0 \leq \text{score}(1, Y) \leq t \\ \Leftrightarrow\ &0 \leq \sqrt{E} + Z_1 \leq t\sqrt{(\sqrt{E} + Z_1)^2 + Z_2^2 + \cdots + Z_n^2}. \\ \Leftrightarrow\ &(1 - t^2)(\sqrt{E} + Z_1)^2 \leq t^2 \left(Z_2^2 + \cdots + Z_n^2\right). \\ \Leftrightarrow\ &\sqrt{1 - t^2}(\sqrt{E} + Z_1) \leq t\sqrt{Z_2^2 + \cdots + Z_n^2}. \\ \Leftrightarrow\ &\sqrt{E} + Z_1 \leq \frac{t}{\sqrt{1 - t^2}}\sqrt{Z_2^2 + \cdots + Z_n^2}. \end{aligned}$$

Finally,

$$\Pr(\text{score}(1, Y) \leq t) = \Pr\left(\sqrt{E} + Z_1 \leq \frac{t}{\sqrt{1 - t^2}}\sqrt{Z_2^2 + \cdots + Z_n^2}\right).$$

(c)

Let $m = \lceil 2^{\rho n} \rceil$ and $E = E_b \log_2 m \geq n\rho E_b$.

We want to show that if $\rho E_b / \sigma^2 > t^2/(1 - t^2)$, then

$$\Pr(\text{score}(1, Y) \leq t) \overset{n\to\infty}{\to} 0.$$

First,

$$\Pr\left(\frac{\sqrt{E}}{\sqrt{n}} + \frac{Z_1}{\sqrt{n}} \leq \frac{t}{\sqrt{1 - t^2}}\sqrt{\frac{Z_2^2 + \cdots + Z_n^2}{n}}\right).$$

We analyze the asymptotic behavior of each term as $n \to \infty$:

1. From the energy bound, $\frac{\sqrt{E}}{\sqrt{n}} \geq \sqrt{\rho E_b}$.
2. $\frac{Z_1}{\sqrt{n}} \xrightarrow{p} 0$ since Z_1 has fixed variance σ^2.
3. By the Law of Large Numbers,

$$\frac{Z_2^2 + \cdots + Z_n^2}{n} \xrightarrow{p} \sigma^2.$$

Thus,

$$\begin{aligned} \frac{\sqrt{E}}{\sqrt{n}} + \frac{Z_1}{\sqrt{n}} \leq \frac{t}{\sqrt{1 - t^2}}\sqrt{\frac{Z_2^2 + \cdots + Z_n^2}{n}} &\overset{n\to\infty}{\to} \sqrt{\rho E_b} \leq \frac{t}{\sqrt{1 - t^2}}\sigma, \\ &\Leftrightarrow \frac{\rho E_b}{\sigma^2} \leq \frac{t^2}{1 - t^2}. \end{aligned}$$

However,

$$\frac{\rho E_b}{\sigma^2} > \frac{t^2}{1 - t^2}.$$

Then,

$$\Pr(\text{score}(1, Y) \leq t) \to 0 \quad \text{as } n \to \infty.$$

(d)

Suppose that the codewords $C_1, \dots, C_m$ are chosen uniformly and independently at random on the n-dimensional sphere of radius $\sqrt{E}$. Message 1 is sent, so $Y = C_1 + Z$.

We want to show that

$$\Pr\left(\max_{2\le i\le m} \text{score}(i, Y) \ge t\right) \le \frac{1}{2}(m-1)(1-t^2)^{(n-1)/2}.$$

First,

$$\left\{\max_{2\le i\le m} \text{score}(i, Y) \ge t\right\} = \bigcup_{i=2}^{m} \{\text{score}(i, Y) \ge t\}.$$

Applying the Union Bound,

$$\Pr\left(\max_{2\le i\le m} \text{score}(i, Y) \ge t\right) \le \sum_{i=2}^{m} \Pr(\text{score}(i, Y) \ge t).$$

Since $C_i \ \forall\ i \ge 2$ is independent of C_1, Z and Y. The vector C_i is uniformly distributed on the sphere of radius $\sqrt{E}$, which means $U_{(i)} = C_i/\sqrt{E}$ is uniformly distributed on the unit sphere.

Thus:

$$\text{score}(i, Y) = \frac{\langle Y, C_i\rangle}{\|C_i\|\|Y\|} = \left\langle \frac{Y}{\|Y\|}, \frac{C_i}{\sqrt{E}} \right\rangle = \langle v, U_{(i)}\rangle,$$

where $v = Y/\|Y\|$ is a unit vector independent of $U_{(i)}$.

From Problem 1(c), the inner product of a uniform random vector on the unit sphere with any fixed or independent unit vector has the same distribution as U_1.

Using the bound derived in Problem 1(b) for $\Pr(U_1 \ge t)$:

$$\Pr(\text{score}(i, Y) \ge t) \le \frac{1}{2}(1-t^2)^{(n-1)/2}.$$

Summing over the $m-1$ identical terms,

$$\Pr\left(\max_{2\le i\le m} \text{score}(i, Y) \ge t\right) \le \frac{1}{2}(m-1)(1-t^2)^{(n-1)/2}.$$

(e)

Let $m = \lceil 2^{\rho n}\rceil$. We want to show that if $(1-t^2) < 2^{-2\rho}$, then $\Pr\left(\max_{2\le i\le m} \text{score}(i, Y) \ge t\right) \to 0$ as $n \to \infty$.

Using $m \le 2^{\rho n} + 1$,

$$\frac{1}{2}(m-1)(1-t^2)^{(n-1)/2} \le \frac{1}{2}2^{\rho n}(1-t^2)^{(n-1)/2} = \frac{1}{2\sqrt{1-t^2}}\left(2^{\rho}\sqrt{1-t^2}\right)^n.$$

For this expression to converge to 0 as $n \to \infty$, the base of the exponential term must be strictly less than 1:

$$2^{\rho}\sqrt{1-t^2} < 1 \implies 2^{2\rho}(1-t^2) < 1 \implies (1-t^2) < 2^{-2\rho}.$$

Then for $(1-t^2) < 2^{-2\rho}$,

$$\Pr\left(\max_{2\le i\le m} \text{score}(i, Y) \ge t\right) \to 0 \quad \text{as } n \to \infty.$$

(f)

We want to show that if $\rho E_b/\sigma^2 > 2^{2\rho} - 1$, then for any $\epsilon > 0$, we can find a code for a large enough n with an average error probability at most ϵ.

First,

$$(1 - t^2) < 2^{-2\rho} \Leftrightarrow 1 - 2^{-2\rho} < t^2 \implies \frac{t^2}{1 - t^2} > \frac{1 - 2^{-2\rho}}{2^{-2\rho}} = 2^{2\rho} - 1.$$

If $\rho E_b/\sigma^2 > 2^{2\rho} - 1$, we can choose a threshold t such as

$$\frac{\rho E_b}{\sigma^2} > \frac{t^2}{1 - t^2} > 2^{2\rho} - 1.$$

Consider the suboptimal decoding rule 'rule-A'. An error occurs under 'rule-A' given message 1 is sent if:

$$\{\text{score}(1, Y) < t\} \cup \left\{ \max_{2 \le i \le m} \text{score}(i, Y) \ge t \right\}.$$

By the Union Bound,

$$\Pr(\text{Error}_{\text{rule-A}}) \le \Pr(\text{score}(1, Y) \le t) + \Pr\left(\max_{2 \le i \le m} \text{score}(i, Y) \ge t \right).$$

- Since $\frac{\rho E_b}{\sigma^2} > \frac{t^2}{1-t^2}$, part (c) guarantees that $\Pr(\text{score}(1, Y) \le t) \to 0$.
- Since $\frac{t^2}{1-t^2} > 2^{2\rho} - 1$, part (e) guarantees that $\Pr\left(\max_{2 \le i \le m} \text{score}(i, Y) \ge t\right) \to 0$.

Thus, $\Pr(\text{Error}_{\text{rule-A}}) \to 0$ as $n \to \infty$. Because the optimal rule in part (a) achieves a smaller or equal error probability than 'rule-A', the average error probability over the ensemble of random codes also tends to 0. By Shannon's random coding argument, since the ensemble average error probability is less than ϵ for large n, there must exist at least one specific deterministic code configuration $c_1, \ldots, c_m$ whose average error probability is at most ϵ.

Problem 3

Let's define the operations

$$T_1 : x \to x, T_2 : (x_1, x_2) \to (-x_2, x_1), T_3 : x \to -x, and T_4 : x \to -T_2(x).$$

For an even positive integer n and $x \in \mathbb{R}^n$, the channel output is

$$Y = T(x) + Z,$$

where $T \in \{T_1, T_2, T_3, T_4\}$ uniformly at random, and $Z \sim \mathcal{N}(0, \sigma^2 I_n)$.

We now solve each part.

(a)

We want to show that if we have $c \in \mathcal{C} \implies -c \in \mathcal{C}$ and the messages are equally likely, then the probability of error is at least 1/2.

Thus, if T_1 is chosen, we have $T_1(c_i) = c_i$. If T_3 is chosen, we would have $T_3(c_i) = -c_i$.

By hypothesis, there exists some index $j \neq i$ such that $c_j = -c_i$.

Therefore,

$$f_{Y|X,T}(y \mid c_i, T_1) = f_{Y|X,T}(y \mid c_j, T_3) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{\|y - c_i\|^2}{2\sigma^2}\right),$$

$$f_{Y|X,T}(y \mid c_i, T_3) = f_{Y|X,T}(y \mid c_j, T_1) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{\|y + c_i\|^2}{2\sigma^2}\right).$$

Since $\Pr(T_1) = \Pr(T_3) = 1/4$, y gives an identical total likelihood for the pairs (c_i, T_1) and (c_j, T_3), as for (c_i, T_3) and (c_j, T_1).

No receiver can distinguish whether c_i was sent via T_1 or c_j was sent via T_3. So, the probability of making a correct decision between these two indistinguishable hypotheses is at most 1/2.

Thus,

$$\Pr(\text{Error}) \geq \frac{1}{2}.$$

(b)

We have,

$$\text{score}(i, y) = \max_{1 \leq k \leq 4} \frac{\langle y, T_k(c_i) \rangle}{\|y\| \|T_k(c_i)\|}.$$

Assume message 1 is sent and transformation T_m is chosen, so $Y = T_m(C_1) + Z$. Consider a suboptimal threshold decoder 'rule-B' with parameter t. An error occurs under 'rule-B' if:

$$\mathcal{E} = \left\{ \max_{1 \leq k \leq 4} \frac{\langle Y, T_k(C_1) \rangle}{\|Y\| \|T_k(C_1)\|} < t \right\} \cup \left\{ \max_{2 \leq i \leq m} \max_{1 \leq k \leq 4} \frac{\langle Y, T_k(C_i) \rangle}{\|Y\| \|T_k(C_i)\|} \geq t \right\}.$$

We bound each term:

1. For the first term, logically,

$$\max_{1 \leq k \leq 4} \frac{\langle Y, T_k(C_1) \rangle}{\|Y\| \|T_k(C_1)\|} \geq \frac{\langle Y, T_m(C_1) \rangle}{\|Y\| \|T_m(C_1)\|}.$$

$$\Leftrightarrow \Pr\left(\max_{1 \leq k \leq 4} \frac{\langle Y, T_k(C_1) \rangle}{\|Y\| \|T_k(C_1)\|} < t \right) \leq \Pr\left(\frac{\langle Y, T_m(C_1) \rangle}{\|Y\| \|T_m(C_1)\|} < t \right).$$

Substituting by $Y = T_m(C_1) + Z$ in the right-hand,

$$\frac{\langle Y, T_m(C_1)\rangle}{\|Y\|\|T_m(C_1)\|} = \frac{\langle T_m(C_1) + Z, T_m(C_1)\rangle}{\|T_m(C_1) + Z\|\|T_m(C_1)\|} = \frac{\langle T_m(C_1), T_m(C_1)\rangle + \langle Z, T_m(C_1)\rangle}{\|T_m(C_1) + Z\|\|T_m(C_1)\|}.$$

Knowing that $\|T_m(C_1)\| = \|C_1\|$, $\langle T_m(C_1), T_m(C_1)\rangle = \langle C_1, C_1\rangle$,

$$= \frac{\langle C_1, C_1\rangle + \langle T_m^{-1}(Z), C_1\rangle}{\|C_1 + T_m^{-1}(Z)\|\|C_1\|} = \frac{\langle C_1 + T_m^{-1}(Z), C_1\rangle}{\|C_1 + T_m^{-1}(Z)\|\|C_1\|}.$$

By the rotational invariance of Z and T_m^{-1} is orthogonal, the transformed noise vector $\tilde{Z} = T_m^{-1}(Z)$ has the exact same distribution as Z, i.e., $\tilde{Z} \sim \mathcal{N}(0, \sigma^2 I_n)$. Similarly, since C_1 is uniformly distributed on the sphere, its distribution is invariant under T_m^{-1}. So,

$$\frac{\langle C_1 + \tilde{Z}, C_1\rangle}{\|C_1 + \tilde{Z}\|\|C_1\|} \overset{d}{=} \text{score}(1, Y)_{\text{Prob 2}}.$$

Thus,

$$\Pr\left(\max_{1\le k\le 4} \frac{\langle Y, T_k(C_1)\rangle}{\|Y\|\|T_k(C_1)\|} < t\right) \le \Pr(\text{score}(1, Y)_{\text{Prob 2}} \le t).$$

2. For the second term, we apply the Union Bound,

$$\Pr\left(\max_{2\le i\le m}\max_{1\le k\le 4} \frac{\langle Y, T_k(C_i)\rangle}{\|Y\|\|T_k(C_i)\|} \ge t\right) \le \sum_{i=2}^{m}\sum_{k=1}^{4} \Pr\left(\frac{\langle Y, T_k(C_i)\rangle}{\|Y\|\|T_k(C_i)\|} \ge t\right).$$

Since C_i is independent of Y, then $T_k(C_i)$is independent of Y. Form Problem 1(b), each of the $4(m-1)$ terms is identically bounded by

$$\frac{1}{2}(1 - t^2)^{(n-1)/2}$$

Combining via the Union Bound,

$$\Pr(\text{Error}_{\text{rule-B}}) \le \Pr(\text{score}(1, Y)_{\text{Prob 2}} \le t) + 4 \cdot \left[\frac{1}{2}(m-1)(1-t^2)^{(n-1)/2}\right].$$

By Problem 2(f),

$$\Pr(\text{Error}_{\text{rule-B}}) \le 4 \cdot \Pr(\text{Error}_{\text{rule-A, Prob 2}}).$$

Since the optimal receiver minimizes the probability of error, its error probability is bounded by at most 4 times the upper bound derived in Problem 2.

(c)

Suppose an oracle provides an estimate $\hat{T}$ of the transformation used, with an error probability $\Pr(\hat{T} \neq T) = \delta_1$.

The receiver can implement the system by compensating for the estimated transformation before decoding. Since $T_1, \ldots, T_4$ correspond to multiplications by $1, j, -1, -j$ in the complex plane, each T_k has a T_k^{-1}.

The receiver computes:

$$\tilde{Y} = \hat{T}^{-1}(Y) = \hat{T}^{-1}(T(x) + Z) = \hat{T}^{-1}(T(x)) + \hat{T}^{-1}(Z).$$

We analyze the system performance:

- Case 1: The oracle is correct ($\hat{T} = T$). This occurs with probability $1 - \delta_1$. Then, $\tilde{Y} = x + \tilde{Z}$, where $\tilde{Z} = T^{-1}(Z) \sim \mathcal{N}(0, \sigma^2 I_n)$. Then, we have an error probability of δ.
- Case 2: The oracle is incorrect ($\hat{T} \neq T$). This occurs with probability δ_1.

Finally,

$$\Pr(\text{Error}) = \Pr(\text{Error} \mid \hat{T} = T) \Pr(\hat{T} = T) + \Pr(\text{Error} \mid \hat{T} \neq T) \Pr(\hat{T} \neq T)$$

Upper bounding $\Pr(\text{Error} \mid \hat{T} \neq T)$ by 1,

$$\leq \Pr(\text{Error} \mid \hat{T} = T) \Pr(\hat{T} = T) + \Pr(\hat{T} \neq T) \leq \delta(1 - \delta_1) + \delta_1 \leq \delta + \delta_1.$$

(d)

We want to show that transmitting a short known sequence x_{TX} of length L allows us to implement such an oracle.

So $x_{\text{TX}} = [A, A, \ldots, A]^T \in \mathbb{R}^L$. Then.

$$Y_{\text{TX}} = T(x_{\text{TX}}) = T_k(x_{\text{TX}}) + Z.$$

Since x_{TX} is known, the receiver can implement a Maximum Likelihood (ML) classifier to estimate T,

$$\hat{T} = \arg \max_{1 \leq k \leq 4} \langle Y_{\text{TX}}, T_k(x_{\text{TX}}) \rangle.$$

Because the four signatures $T_1(x_{\text{TX}}), \ldots, T_4(x_{\text{TX}})$ are mutually orthogonal vectors in $\mathbb{R}^L$ with equal energy LA^2, so we can reduce the problem to noise.

Then,

$$\delta_1 = \Pr(\hat{T} \neq T) \leq 3Q\left(\frac{\sqrt{LA^2}}{\sigma}\right).$$

Since $Q(x) \to 0$ exponentially as $x \to \infty$, we can make δ_1 arbitrarily small by choosing a sufficiently large L or symbol amplitude A, thereby successfully implementing the required oracle.